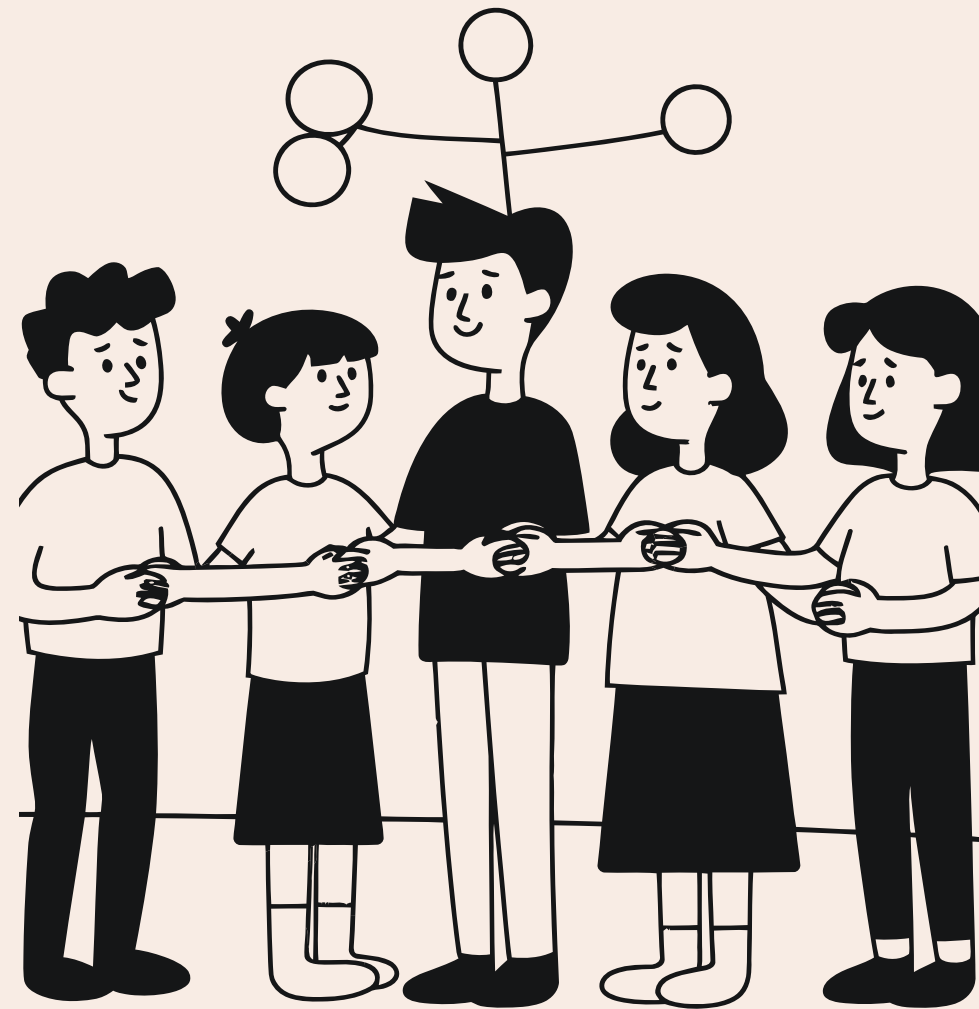
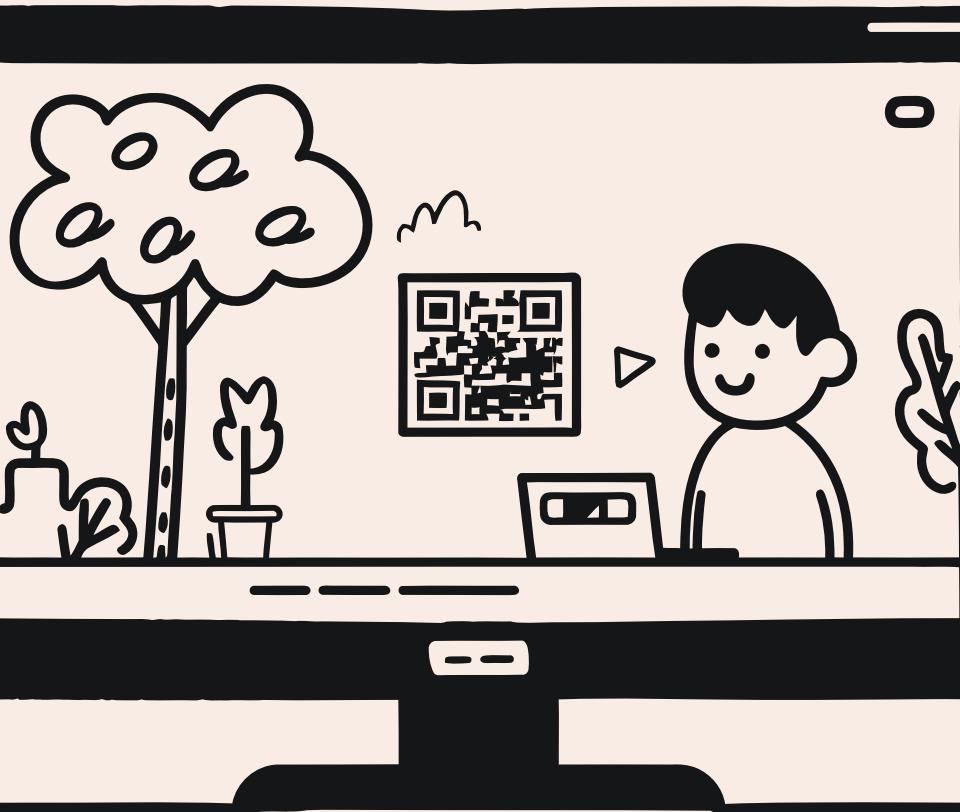


Conversational Knowledge Graph Extraction

Does conversational schema discovery produce better results than a one-shot prompt – and does it generalize?

V3 - 05/06/2026





Demo

Base Case

One-shot prompt on validation emails – expected high precision, limited flexibility.

Interesting Case

5-turn conversation – Location & Transaction entities added, Date & Quantity rejected based on user feedback.



The Problem & Data

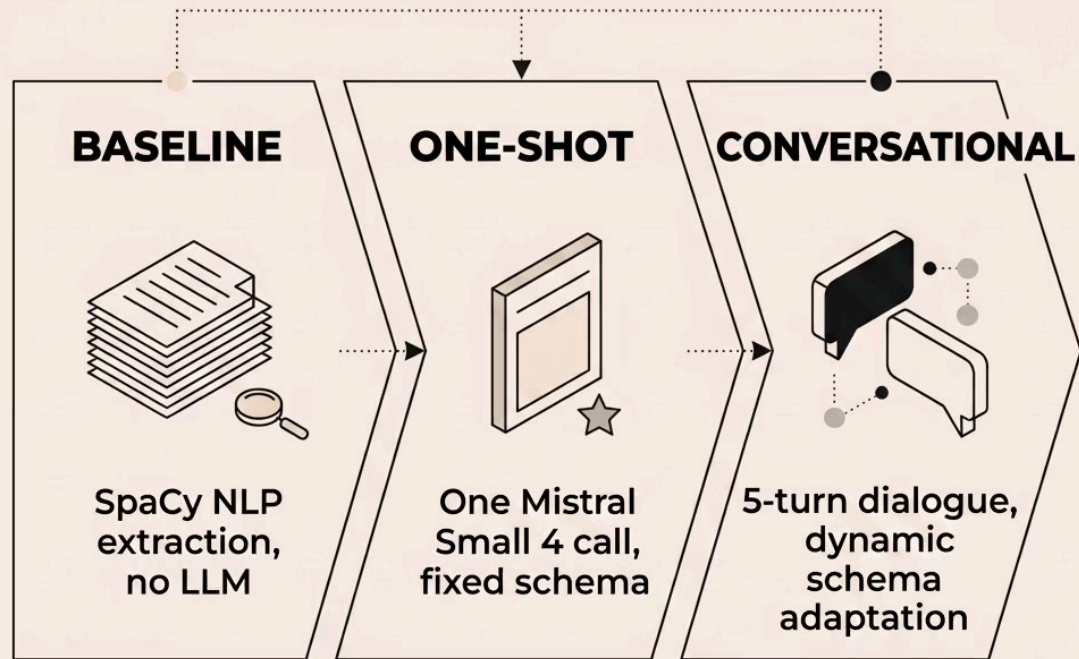
Research Questions

- Does conversational graph extraction outperform a predefined ontology?
- Does the schema generalize – or overfit the conversation?

The Dataset

325 emails from Jeffrey Epstein's personal account (Epstein Files, public domain). 70/30 split: 228 validation, 97 test.

System Architecture



Core Components

- **LLM Engine:** Mistral Small 4 via official APIs (Apache 2.0)
- **Baseline:** SpaCy `en_core_web_trf`
- **Entity Resolution:** Sentence Transformers (`all-MiniLM-L6-v2`, threshold 0.9)
- **Graph Tools:** NetworkX (analysis), Gephi (visualization)

☐ Temperature 0.0, 5 conversational turns, 50 bootstrap samples per condition.



Related Work

Fully Automated KG Construction

AutoSchemaKG – LLMs build KGs without human intervention, but computationally intensive and model-biased.

Expert-in-the-Loop Systems

SCHEMA-MINER – LLM + expert feedback for scientific schema mining. Prompt-sensitive outputs.

The Gap We Fill

No prior work studies KG generation from **free dialogue with a non-expert user** discovering requirements interactively. This is our contribution.

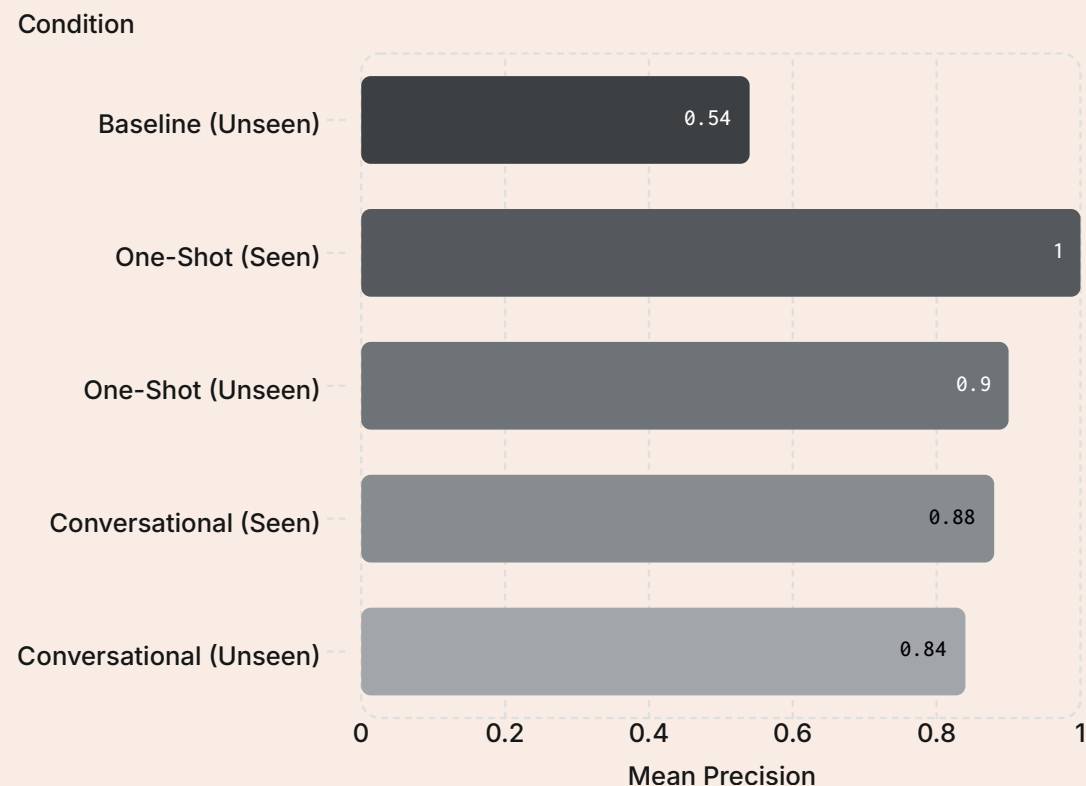
Evaluation Design

Ground Truth

100 hand-labeled triplets (person-relation schema).
Bootstrap resampling on 50 samples per condition.

Metrics

- **Precision** – primary performance metric
- **Generalization Gap** – accuracy drop from validation to test
- **Schema Coverage** – new relation types appearing on test data



Results

0.54

Baseline Precision

SpaCy on unseen data – lowest performer

10%

One-Shot Drop

Precision fell 1.00 → 0.90 on unseen data

4%

Conversational Drop

Only 0.88 → 0.84 – more stable, less overfitting

1.0

Schema Coverage

Both LLM approaches used all 8 relation types



Entity resolution saw 0% reduction (760 → 760 nodes) – Mistral Small 4 handles deduplication well on its own.

What Was Hard



Data Processing

Designing the pipeline from start to finish and making it was the hardest part. While only parts of the pipeline were complete it was not possible to do any evaluation



Schema Sensitivity

Conversational schema quality depends heavily on the user and the specific dialogue – hard to standardize.



Ground Truth Scale

Only 100 hand-labeled triplets. Manual labeling is time-consuming and introduces subjectivity.

Takeaways

→ **Conversational extraction generalizes better**

4% vs 10% precision drop on unseen data – less overfitting than one-shot.

→ **LLMs beat traditional NLP for this task**

Baseline precision 0.54 vs 0.84-0.90 for LLM approaches.

→ **Non-expert dialogue is an open research gap**

Prior work focuses on automation or expert-in-the-loop – free dialogue with non-experts remains understudied.

→ **Scaling Evaluation**

Future work should focus on creating a big ground truth dataset, ingesting more data and refining the evaluation methodology to strengthen statistical confidence in these findings.

📄 Solo project. All work – design, implementation, evaluation, and writing – by the author.